sed on the Final Physical Prototype — Not Fabrication-Validated

because photographed breakout revisions may already contain them.

Main 5 V input

Main power switch

AMS1117-3.3 regulator module

10 uF regulator input

10 uF regulator output

100 nF regulator bypass

SIM800 4.0 V input

GSM VBAT switch

1000 uF low-ESR SIM800 bulk

100 nF SIM800 bypass

SIM800 power header

VK2828U7G5LF GPS module

BTN_MODE_PB12
BTN_UP_PB13
BTN_DOWN_PB14
BTN_NEXT_PB15
BTN_HELP_PA8
GSM_TX_FROM_MCU_PA9
GSM_TX_TO_MCU_RX_PA10
TEMP_DQ_PA11
OLED_SCL_PB6
OLED_SDA_PB7

STM32F103C8T6 Blue Pill socket

10k SIM800 RX divider upper

SIM800 UART / PWRKEY header

47k SIM800 RX divider lower

DS18B20 module

4.7k DS18B20 pull-up (DNP if module fitted)

ADXL345 breakout

4.7k ADXL SDA pull-up (DNP)

MAX30102 breakout

4.7k MAX SDA pull-up (DNP)

0.96 inch OLED I2C module

4.7k OLED SDA pull-up (DNP)

Mode button

Up button

Down button

Next button

Help button

1k buzzer base resistor

10k buzzer base pull-down

S8050 low-side buzzer driver

5 V active buzzer

Final physical prototype governs placement; firmware governs confirmed GPIO
Not Fabrication-Validated
Independent undergraduate project reconstruction

Sheet: /
File: FallDetection_Reconstructed.kicad_sch

Title: STM32 Fall Detection — Reconstructed Final Design

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